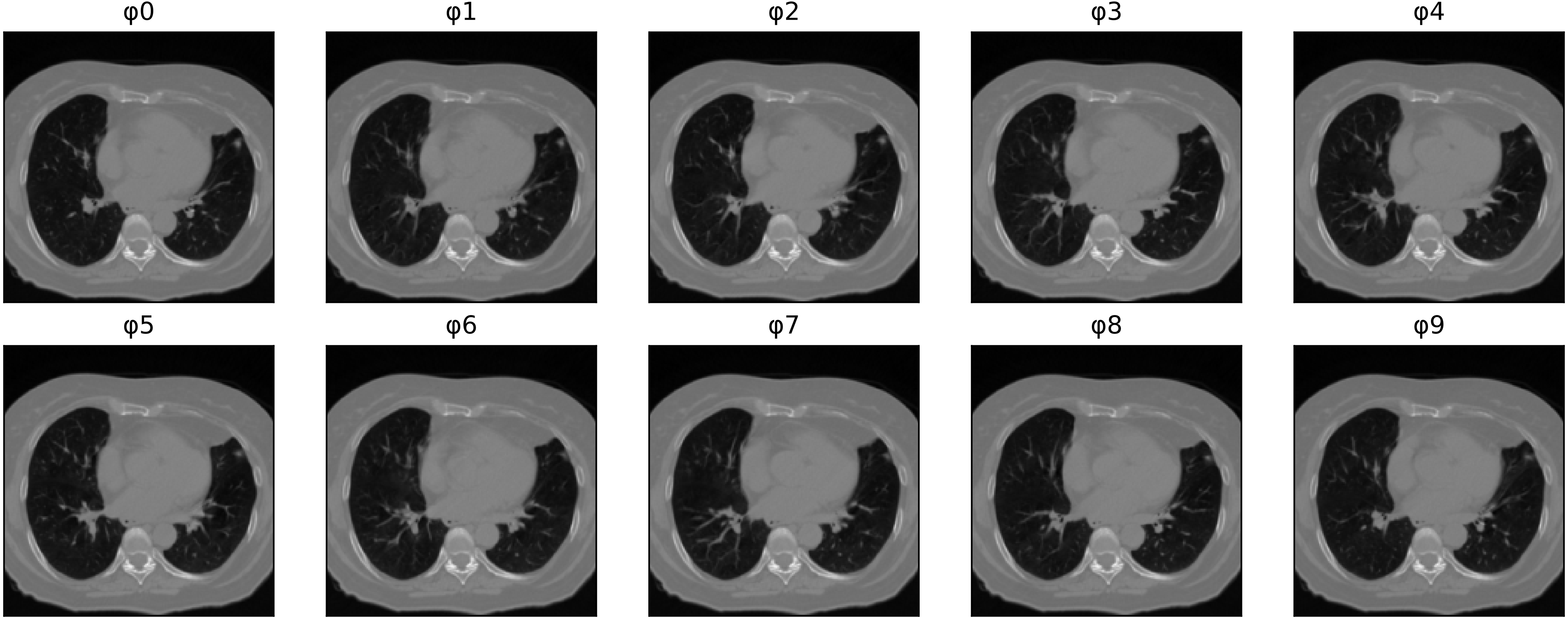
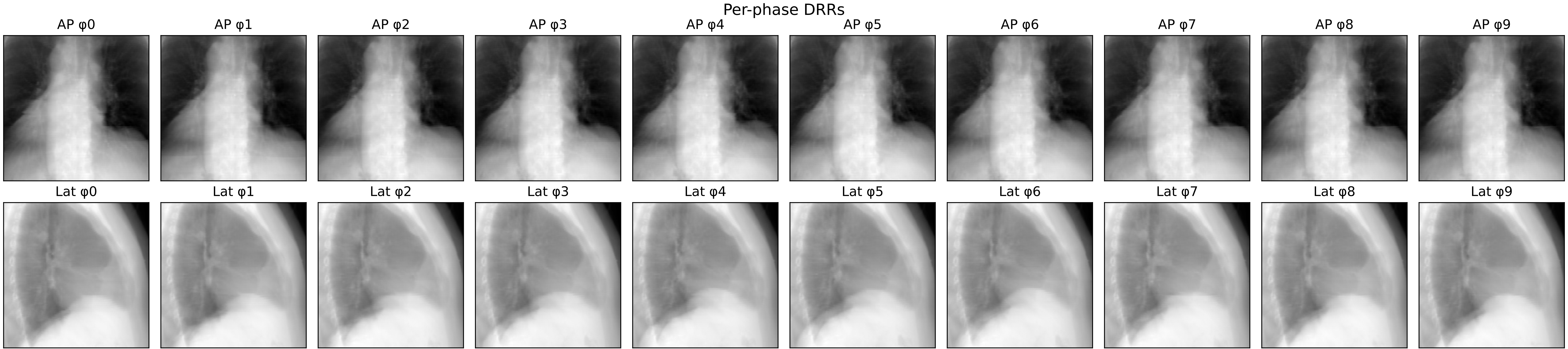
DVF QA pipeline report

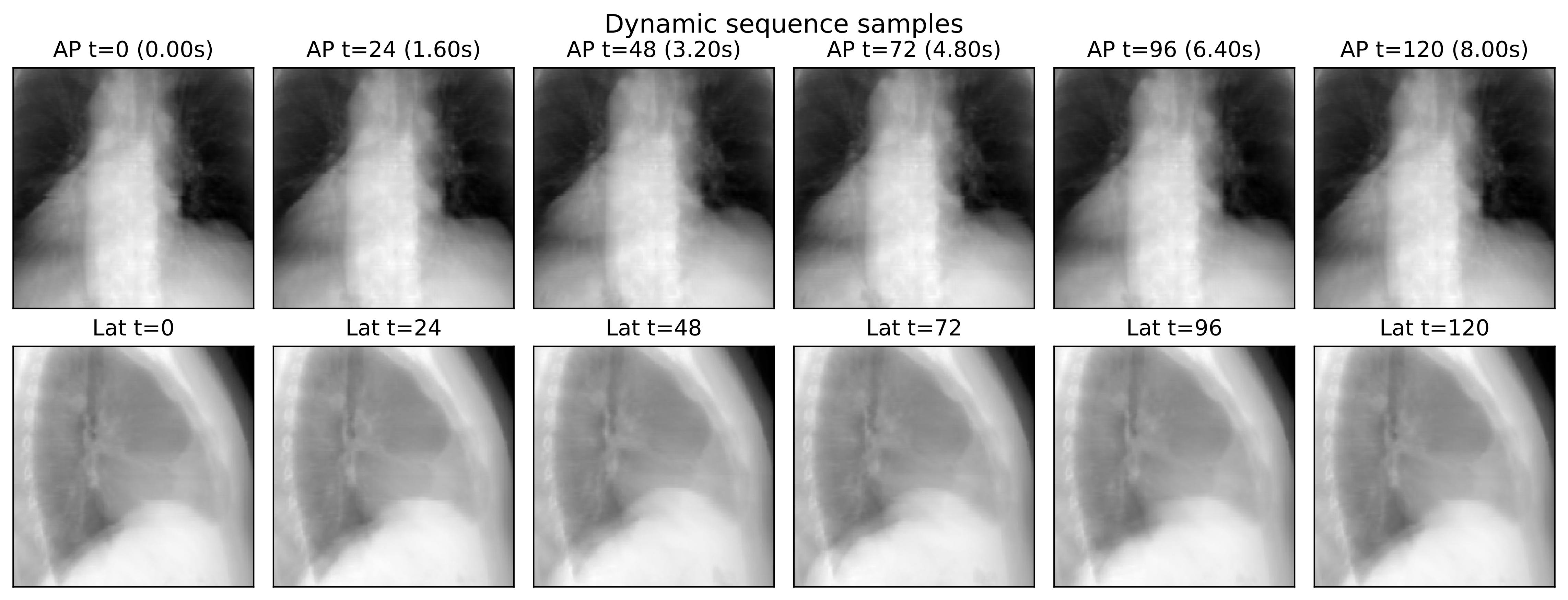
DIR-Lab-case5 [frame_diff]

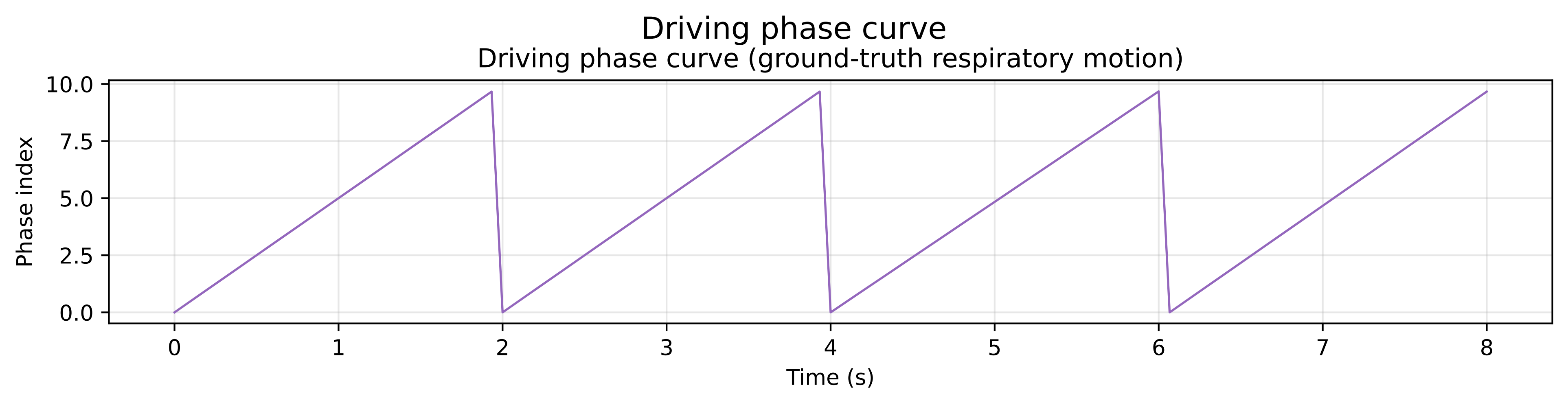
Field	Value
Case	DIR-Lab-case5 [frame_diff]
Generated	2026-05-20 08:03:40
source	Case5Pack
lateral_strategy	frame_diff
fps	15.0
n_cycles	4
frames_per_cycle	30
n_phases	10
ct_shape	(106, 256, 256)
spacing_xyz_mm	(1.1, 1.1, 2.5)
detector_shape	(256, 256)
pixel_spacing_mm	(1.0, 1.0)

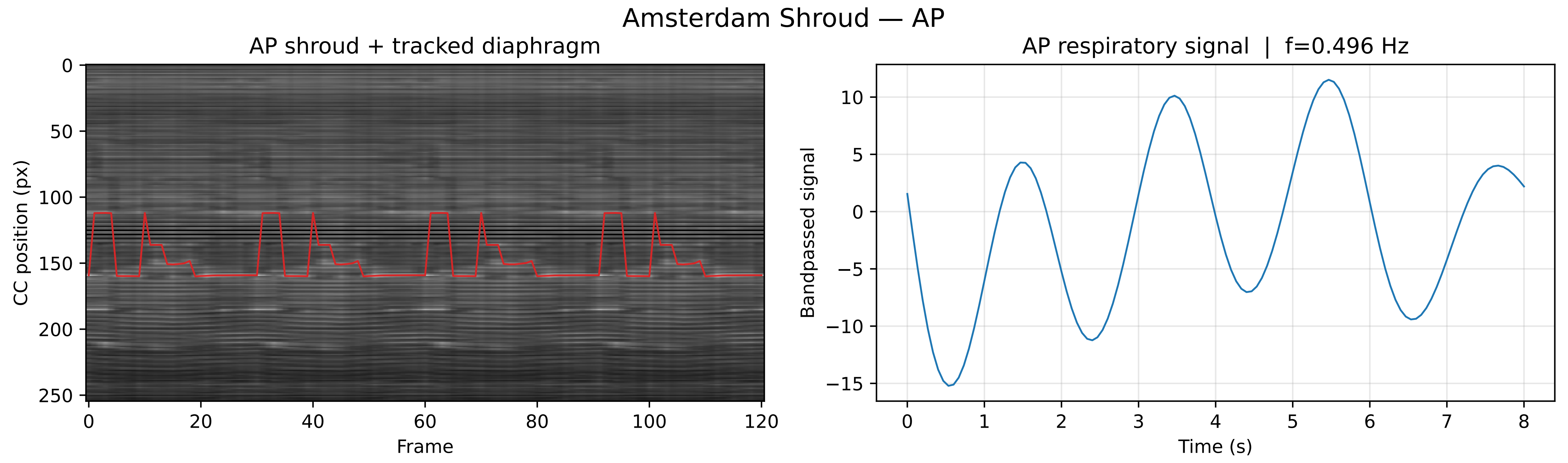
4DCT phase preview



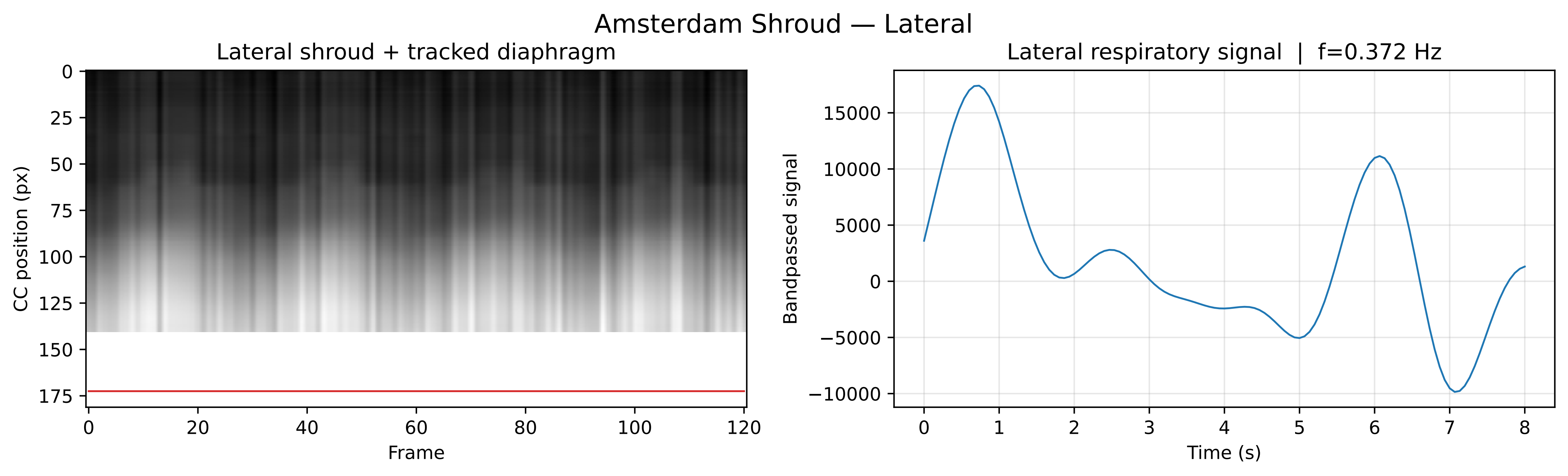

Per-phase DRRs
AP φ0
AP φ1
AP φ2
AP φ3
AP φ4
AP φ5
AP φ6
AP φ7
AP φ8
AP φ9
Lat φ0
Lat φ1
Lat φ2
Lat φ3
Lat φ4
Lat φ5
Lat φ6
Lat φ7
Lat φ8
Lat φ9

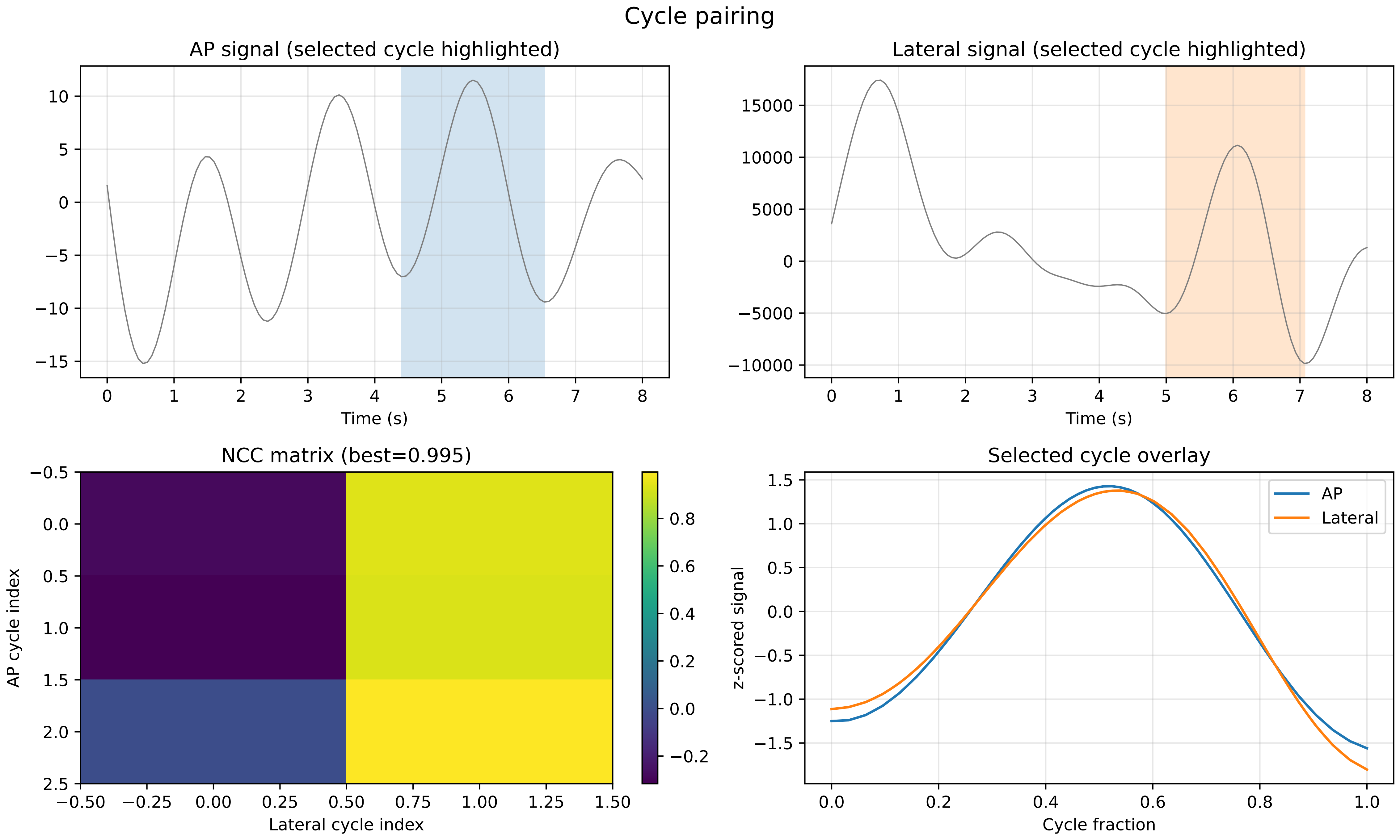

Dynamic sequence samples
AP t=0 (0.00s)
AP t=24 (1.60s)
AP t=48 (3.20s)
AP t=72 (4.80s)
AP t=96 (6.40s)
AP t=120 (8.00s)
Lat t=0
Lat t=24
Lat t=48
Lat t=72
Lat t=96
Lat t=120


Driving phase curve
Driving phase curve (ground-truth respiratory motion)
10.0
7.5
5.0
2.5
0.0
Phase index
0
1
2
3
4
5
6
7
8
Time (s)



Amsterdam Shroud — AP
AP shroud + tracked diaphragm
CC position (px)
Frame
AP respiratory signal | f=0.496 Hz
Bandpassed signal
Time (s)


Amsterdam Shroud — Lateral
Lateral shroud + tracked diaphragm
CC position (px)
Frame
Lateral respiratory signal | f=0.372 Hz
Bandpassed signal
Time (s)



Cycle pairing
AP signal (selected cycle highlighted)
Time (s)
Lateral signal (selected cycle highlighted)
Time (s)
NCC matrix (best=0.995)
AP cycle index
Lateral cycle index
Selected cycle overlay
AP
Lateral
z-scored signal
Cycle fraction

Metrics

Metric	Value
AP dominant frequency (Hz)	0.4959
AP period (frames)	30.25
Lateral dominant frequency (Hz)	0.3719
Lateral period (frames)	40.33
Best pair NCC	0.9951
AP cycle frames	66 - 98
Lateral cycle frames	75 - 106